

2. Linguistics and Phonetics



Nguyen Thi Thu Trang

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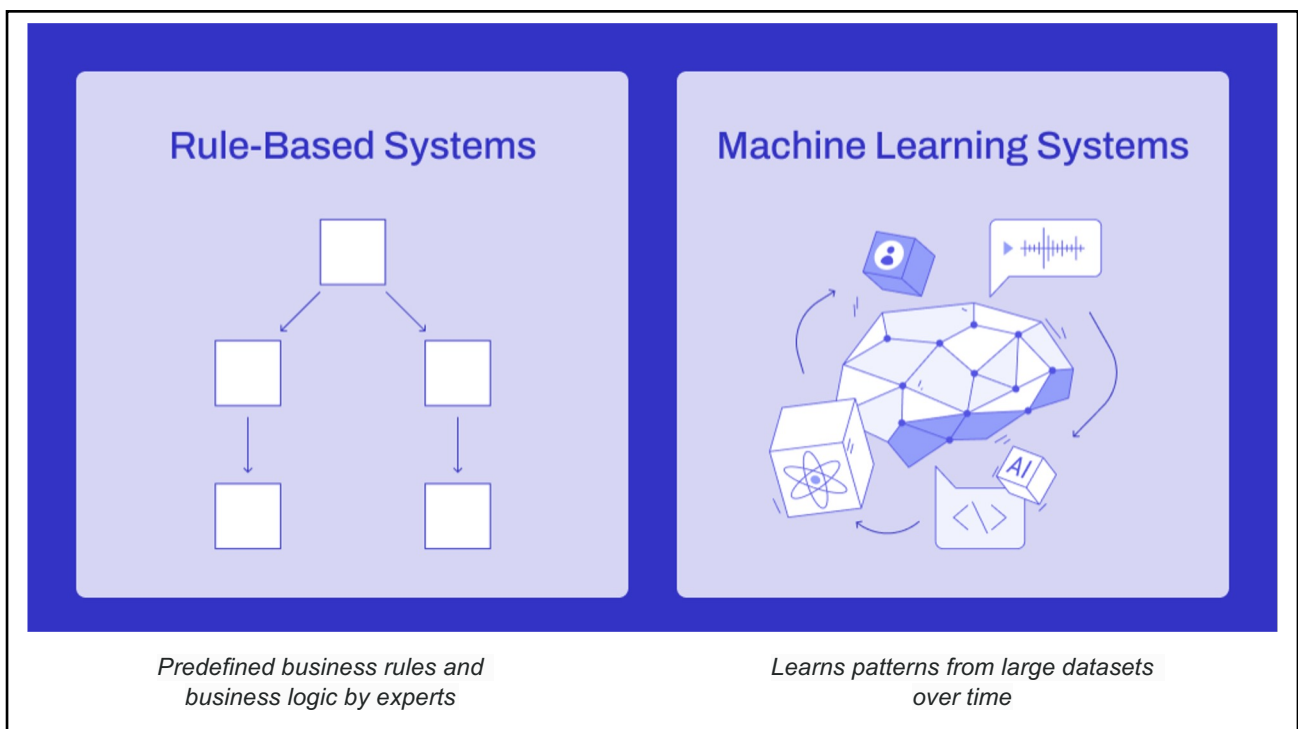
2.1. Basic Phonetics



2

What is the main difference between Rule-based AI and Machine Learning?

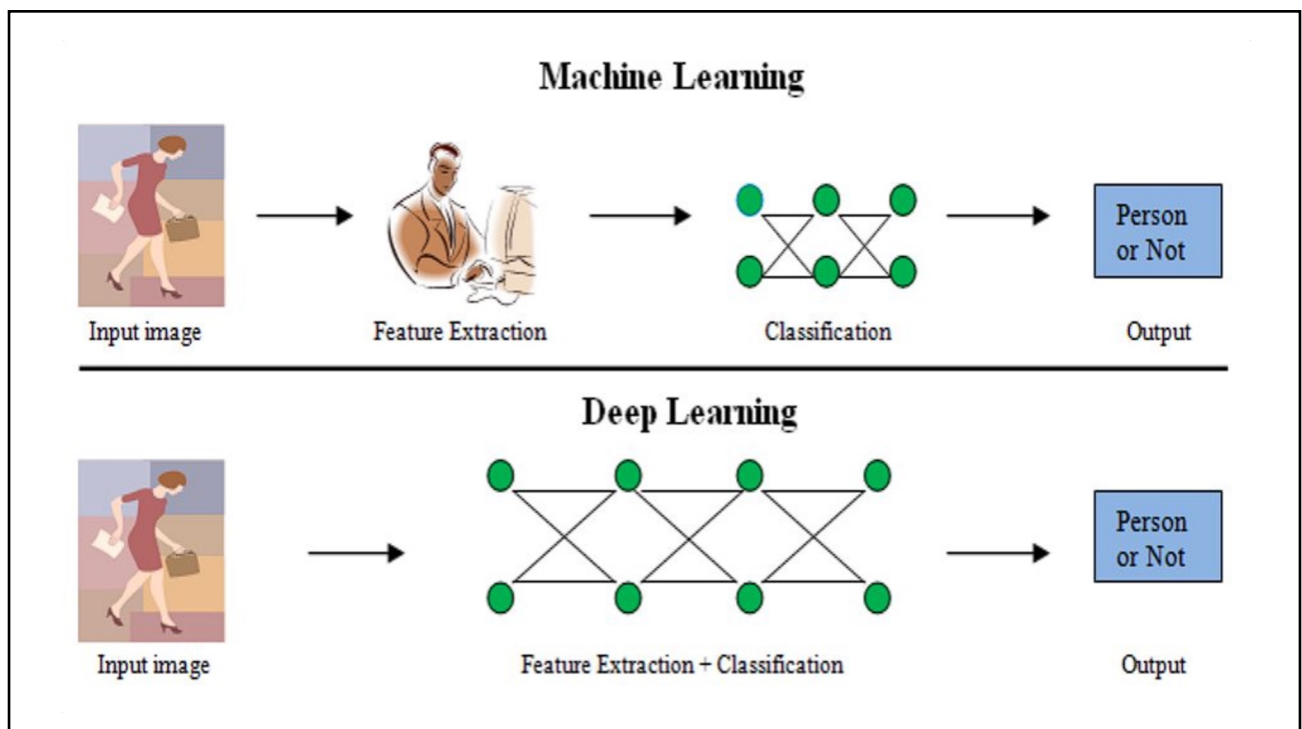
3



4

What is the main difference between Machine Learning and Deep Learning?

5



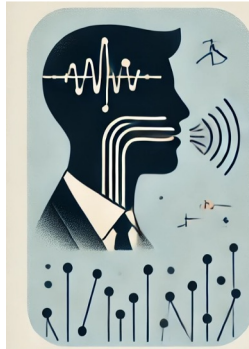
6

Phonology

vs.

Phonetics

- **Sound systems** of a language.
- How **sounds function** in a particular language or languages.
- **Abstract, mental representations of sounds.**



- **Physical production and perception** of speech sounds.
- **Actual acoustic and articulatory properties** of sounds.
- **Universal properties of speech sounds**, not language-specific rules.



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Why Phonology and Phonetics?

Do we need phonology and phonetics to build systems that accurately process spoken language?

- Modern systems (based on deep learning) are far less reliant on encoding phonetic domain knowledge directly than previous approaches
- Allowing deep learning models to learn letter-sound mappings from data can perform much better than hand engineering phonetic structure into a recognition or synthesis system

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Why Phonology and Phonetics?

However ...

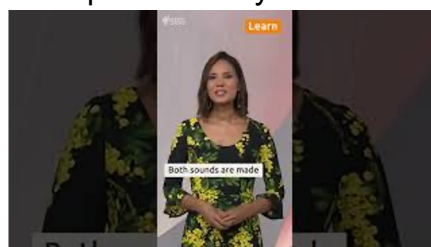
- Basic understanding of phonology and phonetics and speech production helps with describing and debugging spoken language systems
 - E.g. how does an accent/tone change the sound of pronunciations?
- Phonetic categories are not arbitrary. They model the biology of how humans produce speech
 - Understanding the space of possible speech sounds gives a nice perspective on comparing spoken languages across the world, and how they evolve

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What is Phonetics?

- **Definition:**
 - The study of speech sounds, their production, transmission, and perception.
- **Three Main Branches:**
 - **Articulatory Phonetics:** How speech sounds are produced.
 - **Acoustic Phonetics:** Physical properties of speech sounds.
 - **Auditory Phonetics:** How speech sounds are perceived by the listener

Example: Different between /b/ and /p/ sounds



10

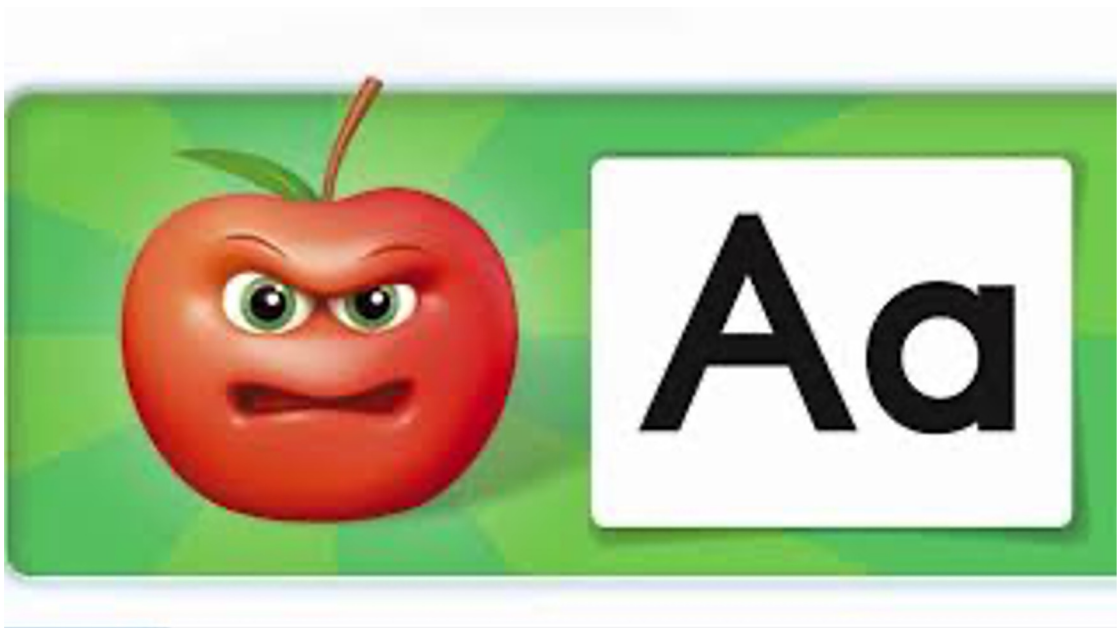
Which parts of the body are involved in producing speech?

Join at
slido.com
#1782 805



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Letter vs. Sound



12

Grapheme vs. Phoneme

- **Grapheme**

The smallest written unit representing a sound in speech.

Examples: "d", "th", "ngh", "a", "au".

- **Phoneme**

The smallest unit of sound in speech.

Examples: /d/, /th/, /ŋ/, /a/, /əʊ/.

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IPA (International Phonetic Alphabet)

<

leave /li:v/
manly /'mænli/

14

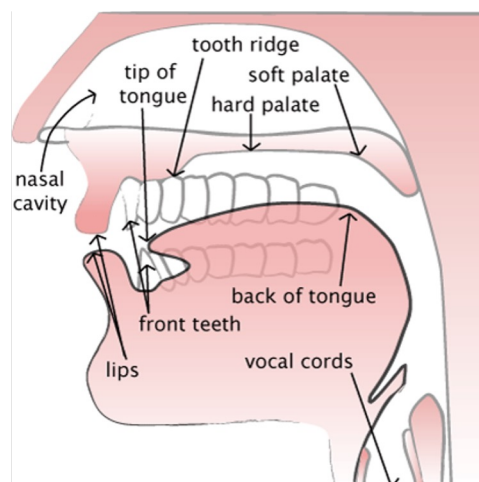
Speech Production

- Flow: we (normally) speak while breathing out. “Pulmonic egressive airstream” (Luồng khí từ phổi)
 - Airstream sets vocal folds in motion. Vibration of vocal folds (rung dây thanh âm) produces sounds.
- Resonance (Điều chỉnh âm thanh): shape of vocal tract causing harmonics
- Articulation (Phát âm): manipulation of airflow
 - Oral tract (Đường miệng): uvula (lưỡi gà), soft palate (velum - vòm miệng mềm), hard palate, tongue, lips, teeth
 - Nasal tract (Đường mũi)

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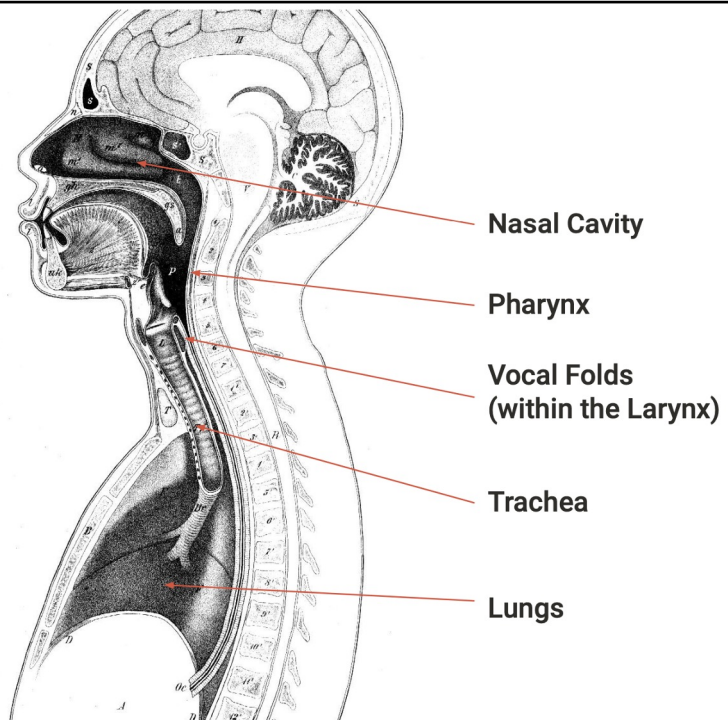
Làm thế nào để tạo ra tiếng nói?

- Hít vào và chuẩn bị không khí: **Phổi** hít vào và điều chỉnh luồng khí qua **cơ hoành (diaphragm)**.
- Đẩy không khí qua **thanh quản**: **Dây thanh âm** rung tạo âm thô.
- Hình thành âm thanh: **Lưỡi, môi** và **khoang miệng** định hình âm thanh.
- Điều chỉnh âm rõ ràng: Phối hợp lưỡi, môi, **răng** và **hàm** để phát âm từ ngữ có ý nghĩa.



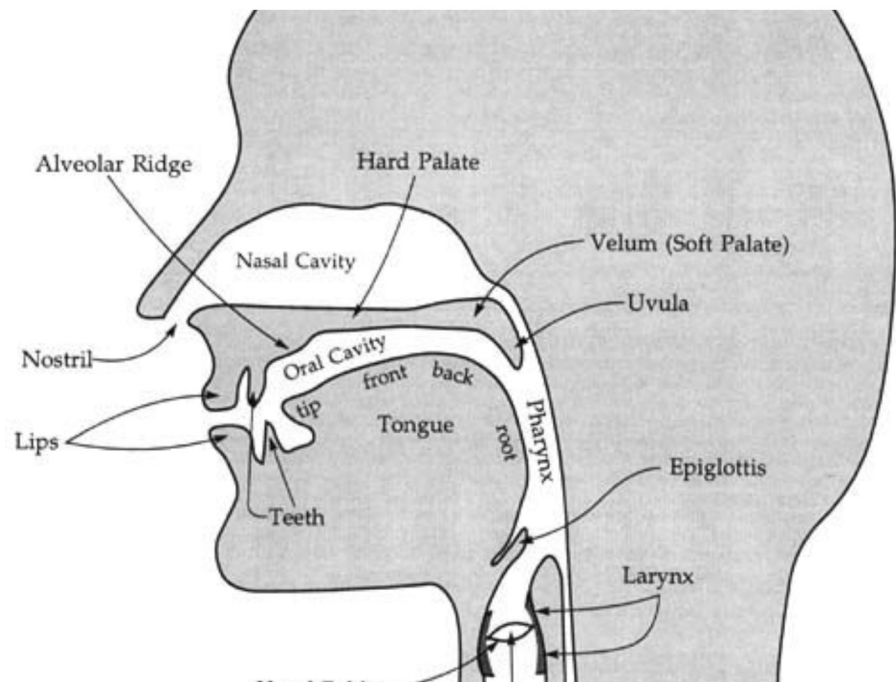
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Sagittal section of the vocal tract



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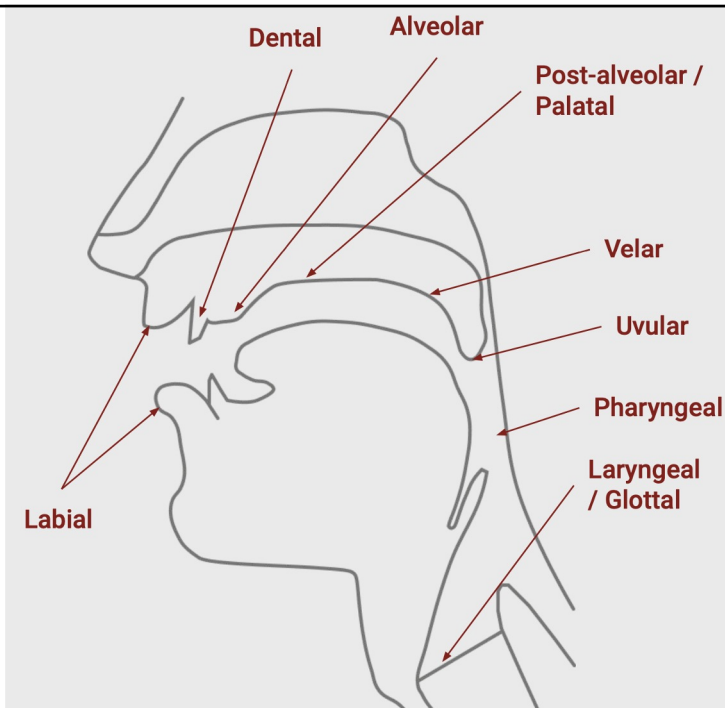
Sagittal section of the vocal tract



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Place of Articulation

- **Consonants** are classified according to the location where the airflow is most constricted
 - This is called **place of articulation**
- Three major kinds of place articulation:
 - **Labial** (with lips)
 - **Coronal** (using tip or blade of tongue)
 - **Dorsal** (using back of tongue)



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THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

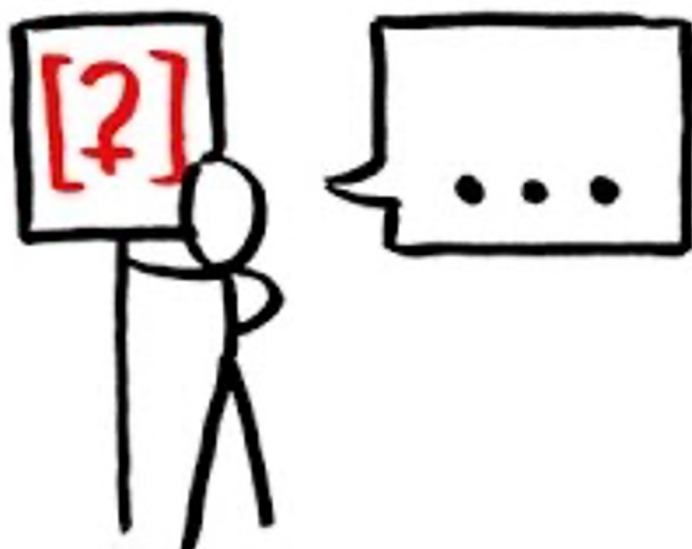
CONSONANTS (PULMONIC)											© 2020 IPA
	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

Place of Articulation

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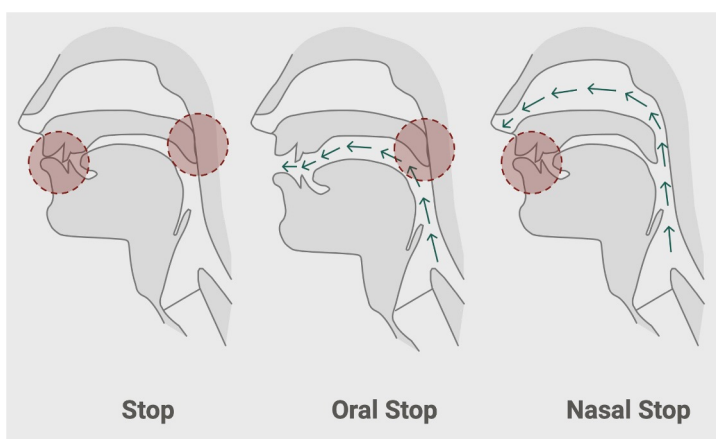
Place of Articulation



21

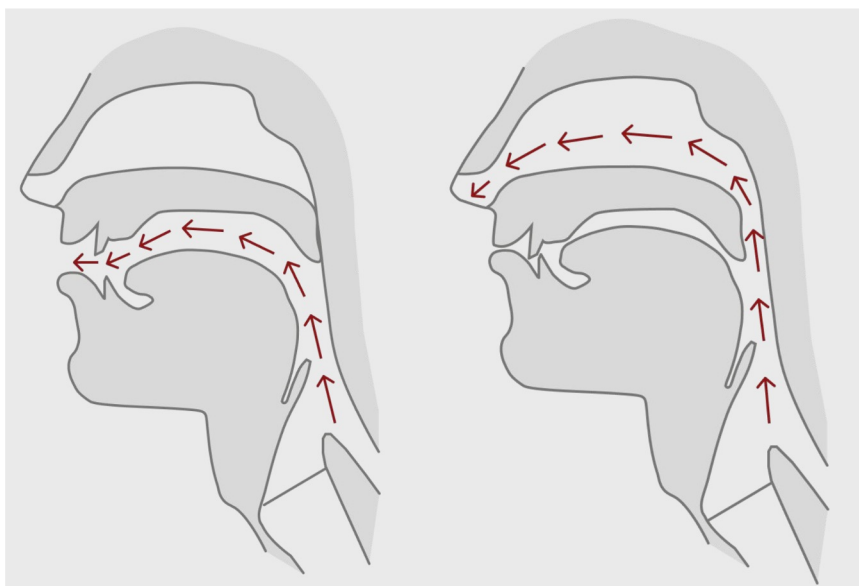
Manner of Articulation

- **Stop (Âm tắc):** complete closure of articulators, so **no air** escapes through mouth
- **Oral stop (Âm tắc miệng):** palate is raised, **no air** escapes through nose. Air pressure builds up behind closure, explodes when released
 - p, t, k, b, d, g
- **Nasal stop (Âm tắc mũi):** oral closure, but palate is lowered, **air escapes through nose**
 - m, n, ng



22

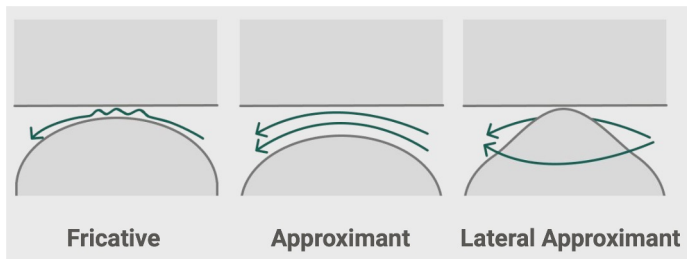
Oral vs. Nasal Sounds



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Articulation Manner of Consonants

- **Fricatives (Âm xát):** close approximation of two articulators, resulting in turbulent airflow between them, producing a hissing sound
 - f, v, s, z, th, dh
- **Approximant (Âm xấp):** not quite-so-close approximation of two articulators, so no turbulence
 - y, r
- **Lateral approximant (Âm xấp bên):** obstruction of airstream along center of oral tract, with opening around sides of tongue
 - l



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Manner of Articulation



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THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

CONSONANTS (PULMONIC)

© 2020 IPA

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Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

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Manner of Articulation

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Voiced vs. Unvoiced Consonants

Presence or absence of vocal cord vibration during the articulation of consonants.

- Vocal cords **vibrate** during pronunciation.
- Air from the lungs causes the vocal cords to rapidly open and close, producing sound waves.
- **Examples:**
 - **/b/** as in *bat*: vocal cords vibrate.
 - **/d/** as in *dog*.
 - **/z/** as in *zoo*.
 - **/v/** as in *van*.
- Vocal cords **do not vibrate** during pronunciation.
- Sound is produced solely by the airflow and contact between the articulators without vocal cord vibration.
- **Examples:**
 - **/p/** as in *pen*: vocal cords do not vibrate.
 - **/t/** as in *top*.
 - **/s/** as in *snake*.
 - **/f/** as in *fish*.

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Voiced and Unvoiced Consonants



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THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

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© 2020 IPA

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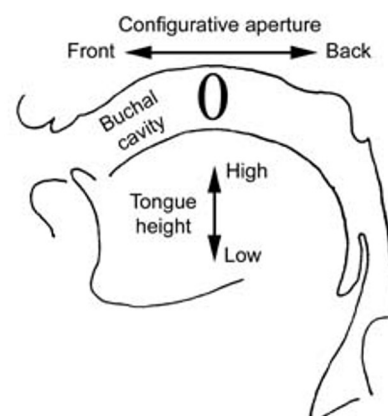
Unvoiced: Left, Voiced: Right

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Tongue Position of Vowels: Tongue Height

How high or low the tongue is in the mouth when producing a vowel:

- **High vowels:** The tongue is raised close to the roof of the mouth.
 - Examples: /i/ (*see*), /u/ (*blue*).
- **Mid vowels:** The tongue is positioned midway between high and low.
 - Examples: /e/ (*bed*), /o/ (*go*).
- **Low vowels:** The tongue is lowered towards the bottom of the mouth.
 - Examples: /æ/ (*cat*), /ɑ/ (*father*).

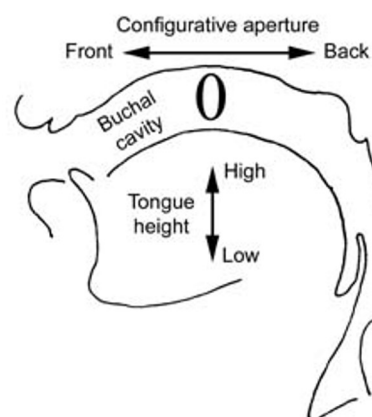


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Tongue Position of Vowels: Tongue Backness

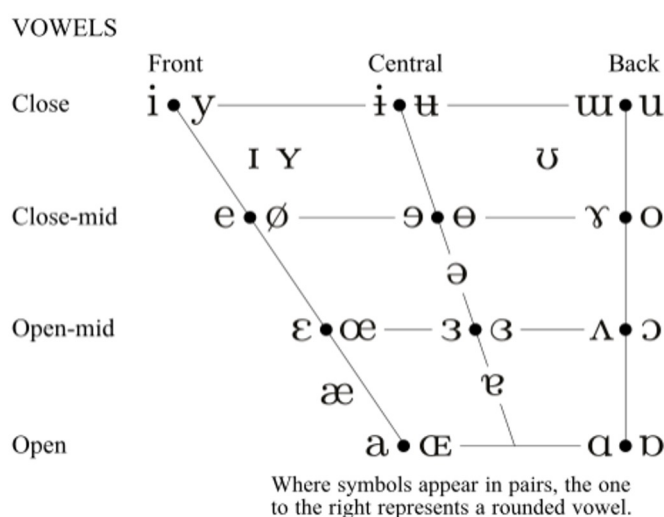
How far forward or backward the tongue is in the mouth:

- **Front vowels:** The tongue is positioned towards the front of the mouth.
 - Examples: /i/ (see), /e/ (bed).
- **Central vowels:** The tongue is centrally positioned.
 - Examples: /ə/ (about), /ʌ/ (cup).
- **Back vowels:** The tongue is positioned towards the back of the mouth.
 - Examples: /u/ (blue), /o/ (go).



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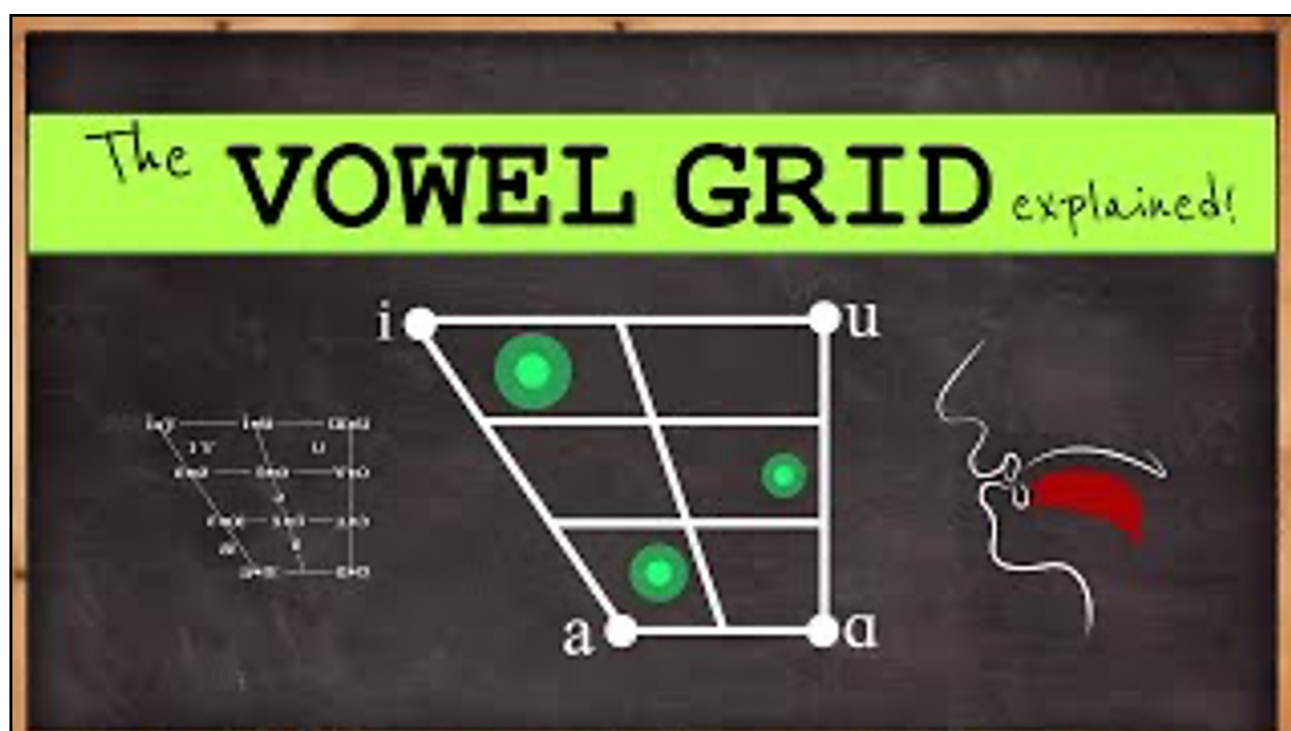
THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)



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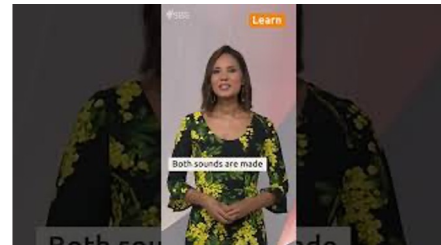


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What is Phonetics?

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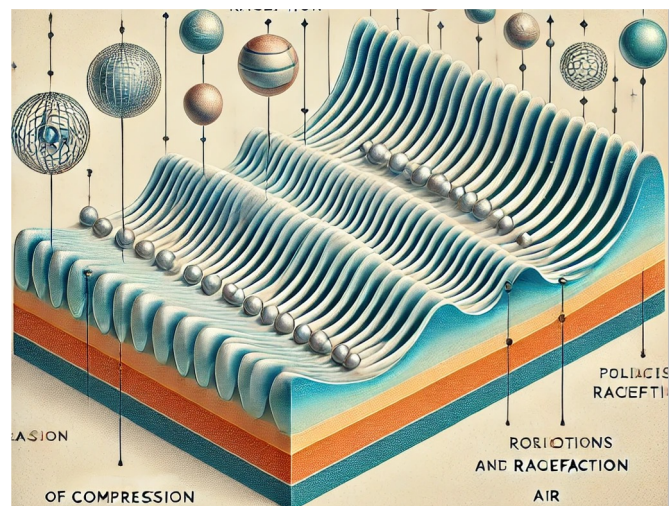
Example: Different between /b/ and /p/ sounds



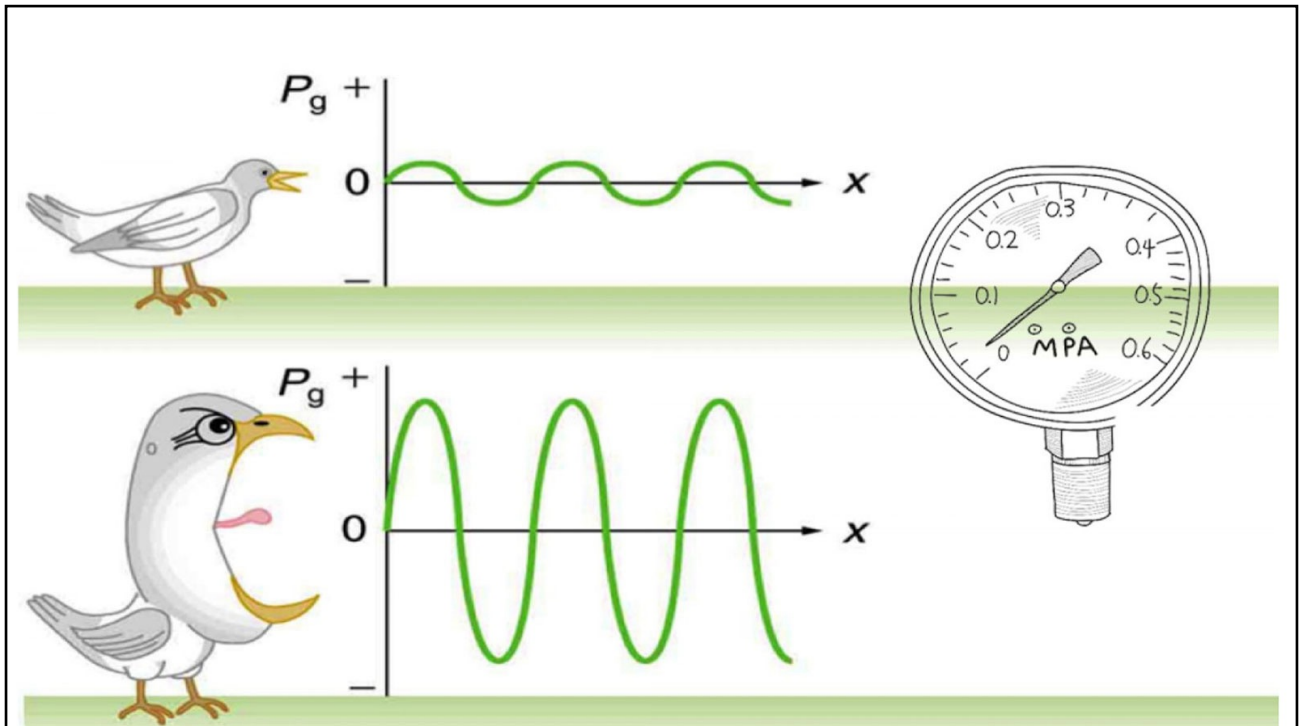
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Sound Waves are Longitudinal Waves

- Particles of the medium vibrate parallel to the direction of the wave's propagation
 - e.g. Sound waves, P-waves, pressure waves....
- Repeating patterns described by **wavelength**, **frequency**, **amplitude**, and **speed**



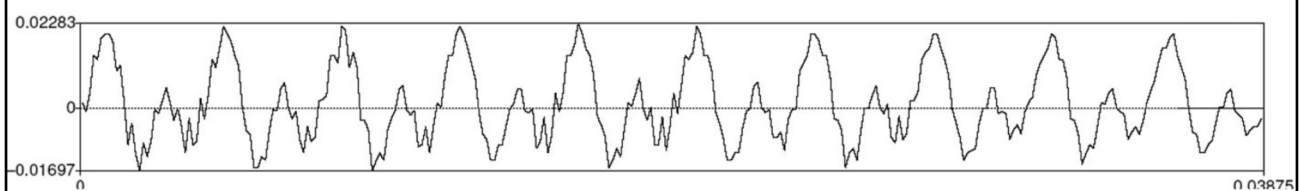
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Back to Waves: Fundamental Frequency

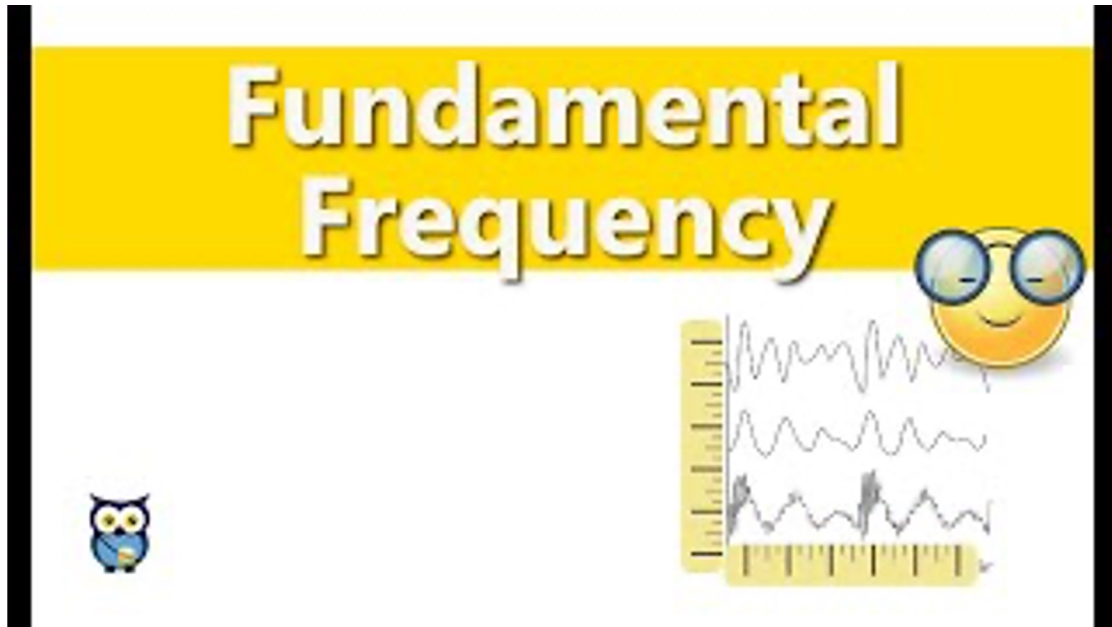
- Waveform of the vowel /iy/



- This is speed that vocal folds move, hence voicing
 - Each peak corresponds to an opening of the vocal folds
 - The low frequency of the complex wave is called the fundamental frequency of the wave or F_0
- Frequency = ?
 → $F_0 = 10 \text{ repetitions} / 0.03875 \text{ seconds} = 258 \text{ Hz}$

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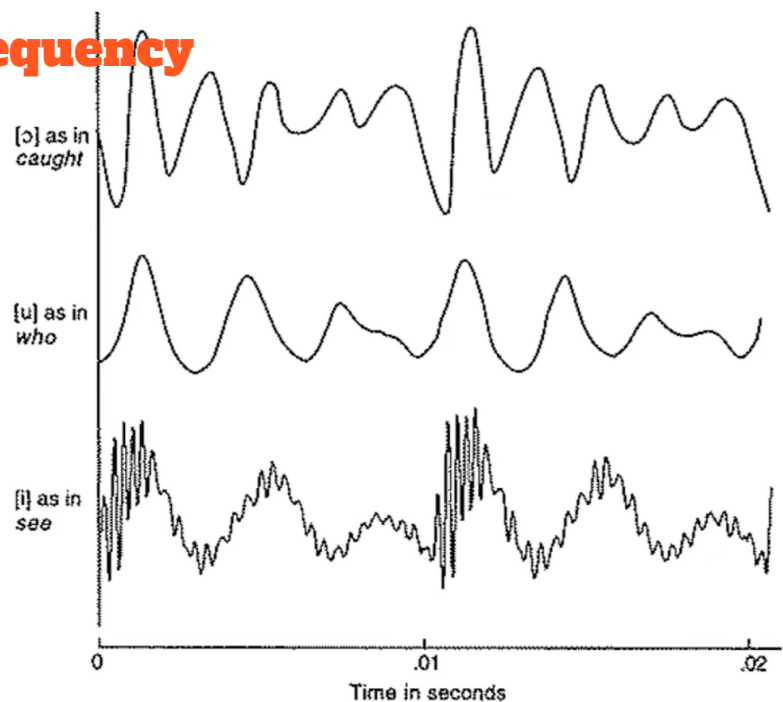
Fundamental Frequency



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Fundamental Frequency

- **Fundamental frequency:** The lowest frequency produced by any particular instrument is known as the fundamental frequency
- **Harmonic:** A harmonic is a waveform whose frequency is an integral multiple of the fundamental frequency
- **E.g.**
 - $f_0 = 1^{\text{st}} \text{ harmonic} = f_1$
 - $f_2 = 2^{\text{nd}} \text{ harmonic} = 2 \times f_0$
 - ...

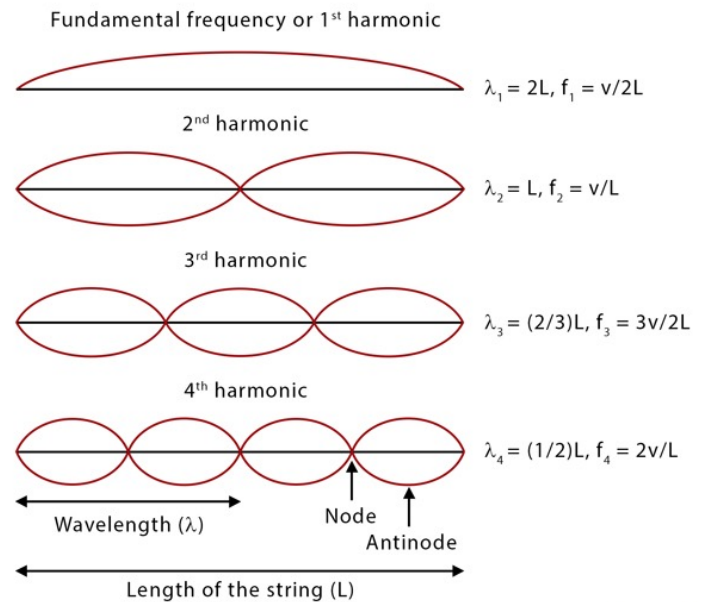


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F0 & Harmonics

- **Node:** A point along a standing wave where the wave has a minimum amplitude
- **Antinode:** A point along a standing wave where the wave has a maximum amplitude

Fundamental Frequency and Harmonics in a String



Source: Physics

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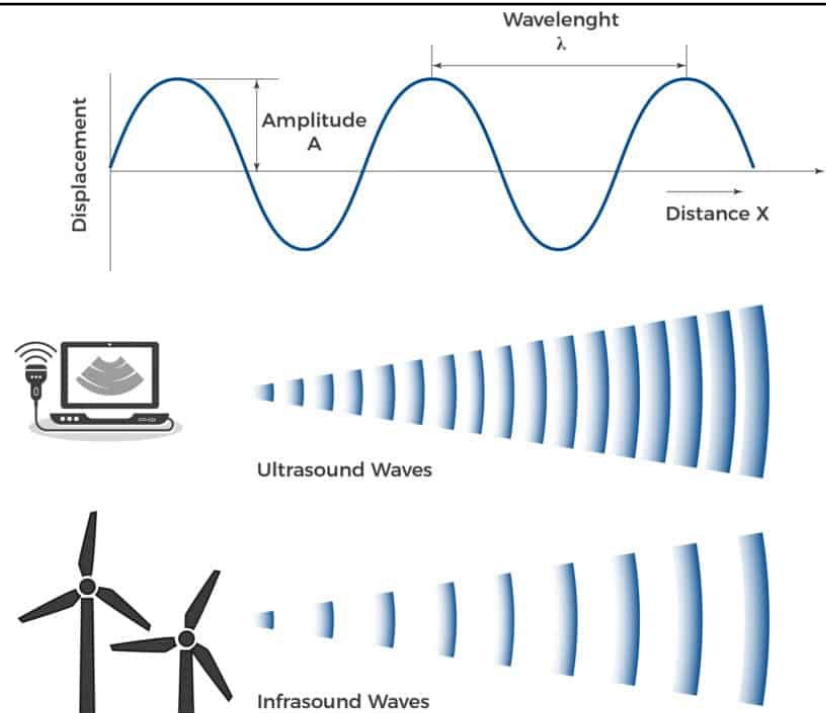
Fundamental Frequency Ranges

Age [years]	Female Frequency [Hz]	Male Frequency [Hz]
Infant	440-590	440-590
3	255-360	255-360
8	215-300	210-295
12	200-280	195-275
15	185-260	135-205
Adults	175-245	105-160

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Amplitude

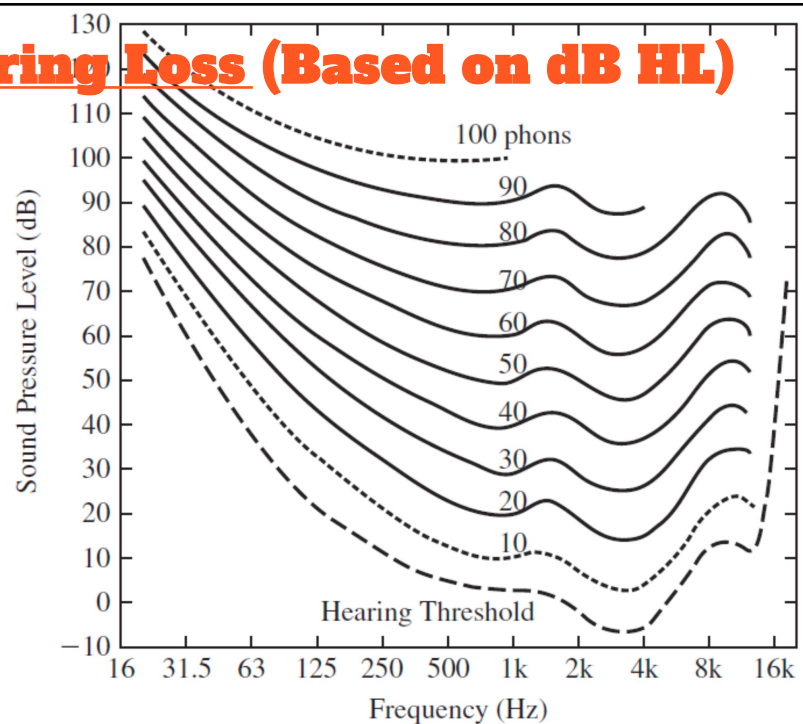
- ✓ Represents the **maximum displacement of particles** from their equilibrium position due to the passage of the wave.
- ✓ Related to the loudness or volume of the sound, with larger amplitudes producing louder sounds.



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Degree of Hearing Loss (Based on dB HL)

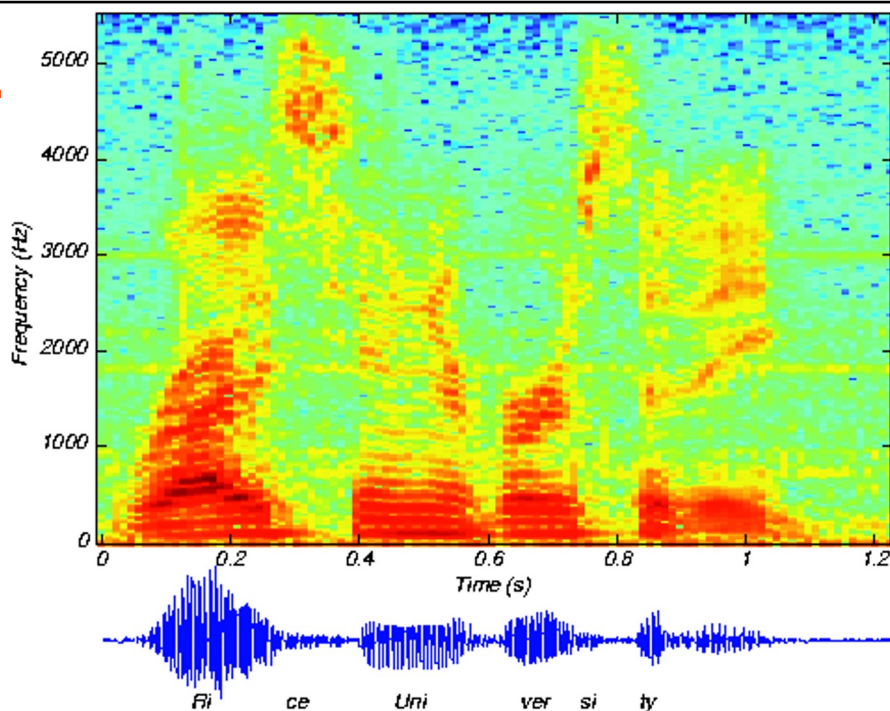
- ✓ **Normal:** -10 to 15 dB
- ✓ **Slight:** 16 to 25 dB
- ✓ **Mild:** 26 to 40 dB
- ✓ **Moderate:** 41 to 55 dB
- ✓ **Moderately Severe:** 56 to 70 dB
- ✓ **Severe:** 71 to 90 dB
- ✓ **Profound:** 90+ dB



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Spectrogram (Ảnh phổ)

A graph that displays the **strength** of a **signal** over **time** for a given **frequency** range



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Waveforms and Spectrograms

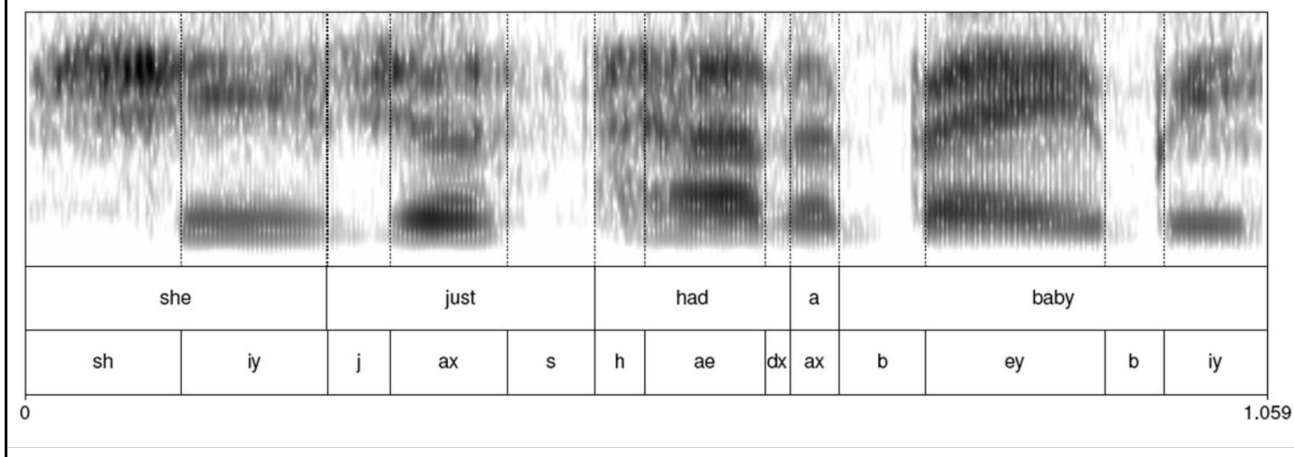
Waveforms and Spectrograms



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Spectrogram: Spectrum + Time Dimension

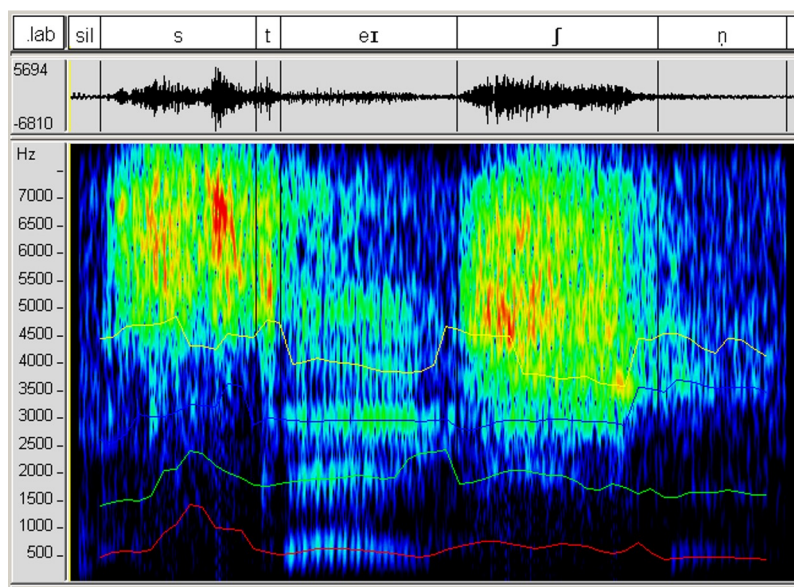
SHE JUST HAD A BABY



47

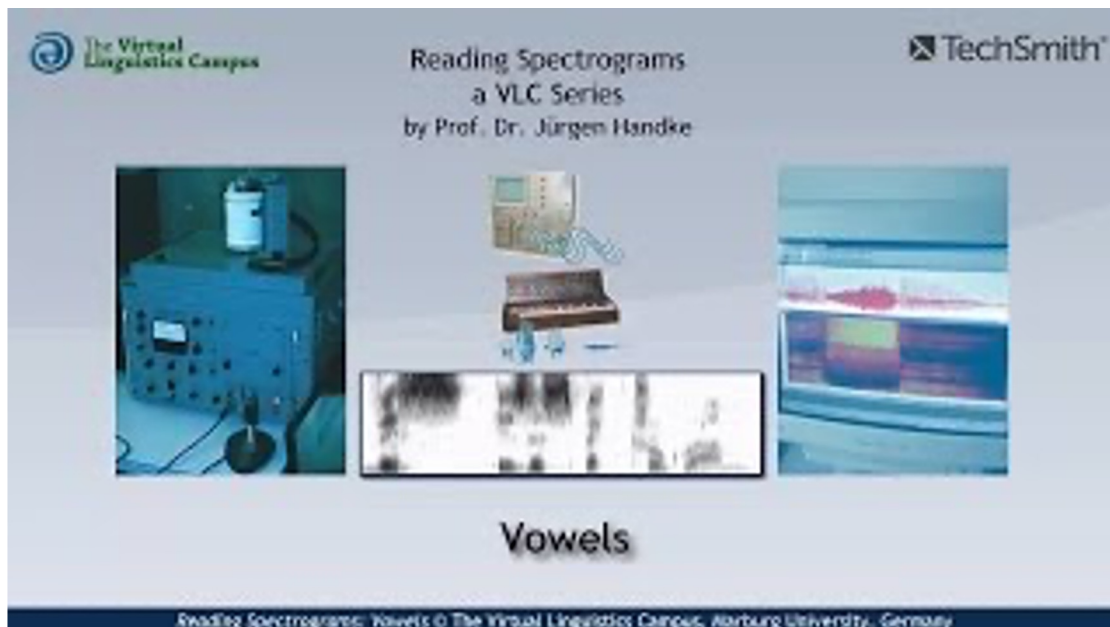
Consonant and Vowel Spectrograms

- **Vowels** have **full formant patterns**, based mostly on F1 and F2
- **Consonants**: No **clear formant patterns** themselves, but rather affect the patterns of the vowels surrounding them



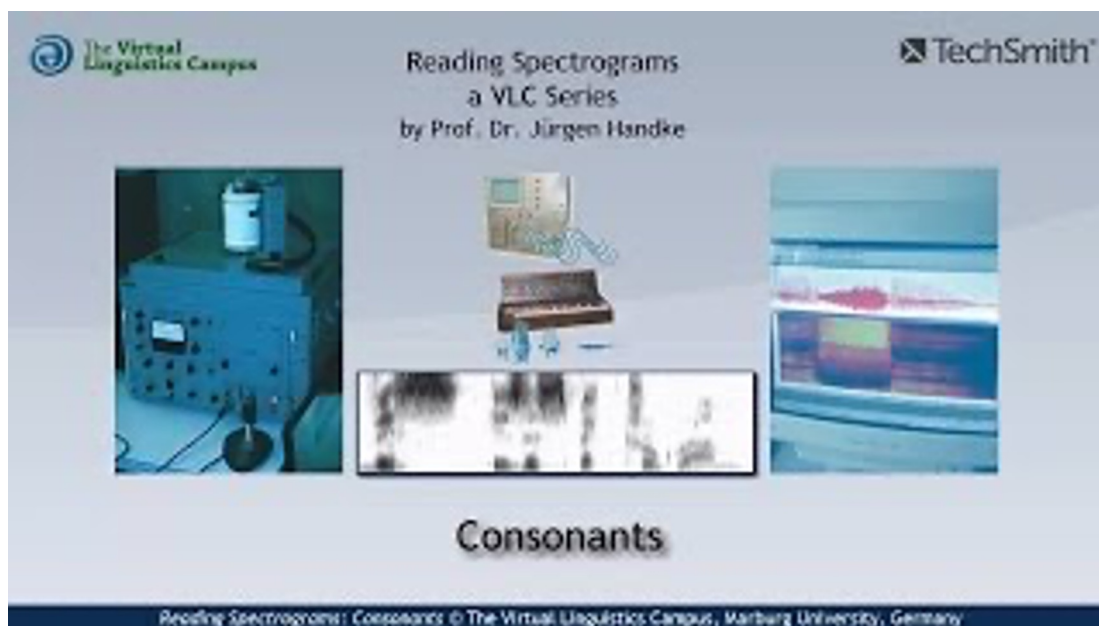
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How to read vowel spectrograms?



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How to read consonant spectrograms?



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Review: Grapheme vs. Phoneme

- **Grapheme**

The smallest written unit representing a sound in speech.

Examples: "d", "th", "ngh", "a", "au".

- **Phoneme**

The smallest unit of sound in speech.

Examples: /d/, /th/, /ŋ/, /a/, /əʊ/.

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Review: IPA (International Phonetic Alphabet)

BẢNG PHIÊN ÂM TIẾNG ANH IPA		monophthongs				diphthongs		Phonemic Chart voiced unvoiced	
		i:	ɪ	ʊ	u:	ɪə	eɪ		
		leave	live	put	pool	year	paper		
		e	ə	ɜ:	ɔ:	ʊə	ɔɪ		
VOWELS		pen	a	girl	door	tourist	boy	Oʊ(əʊ)	
		phone							
		æ	ʌ	ɑ:	ɒ	eə	aɪ	aʊ	
		man	cup	car	hot	chair	fine	house	
CONSONANTS		p	b	t	d	tʃ	dʒ	k	g
		pen	baby	table	door	chip	January	key	girl
		f	v	θ	ð	s	z	ʃ	ʒ
		fan	van	think	this	sun	zoo	shoes	vision
		m	n	ŋ	h	l	r	w	j
		mouth	nose	ring	hear	letter	rain	window	yellow

leave /li:v/
manly /'mænli/

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Exercise

- Download Praat: www.praat.org
- Record the sentence: “She just had a baby”
- Create 2 tiers with TextGrid for annotating for the word (grapheme) and phoneme (using IPA) levels

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Popular packages for signal analysis

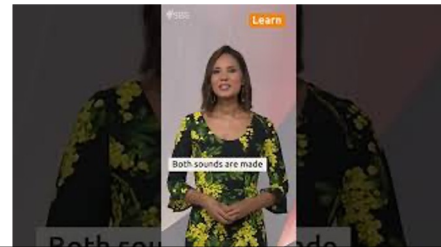
- Praat (www.praat.org)
 - by Paul Boersma & David Weenink, Phonetics Amsterdam
- Wavesurfer (www.speech.kth.se/wavesurfer/)
 - by scientists at Stokholm Technical University - KTH
- Goldwave (www.goldwave.de)
 - commercial program, but available free for trial use

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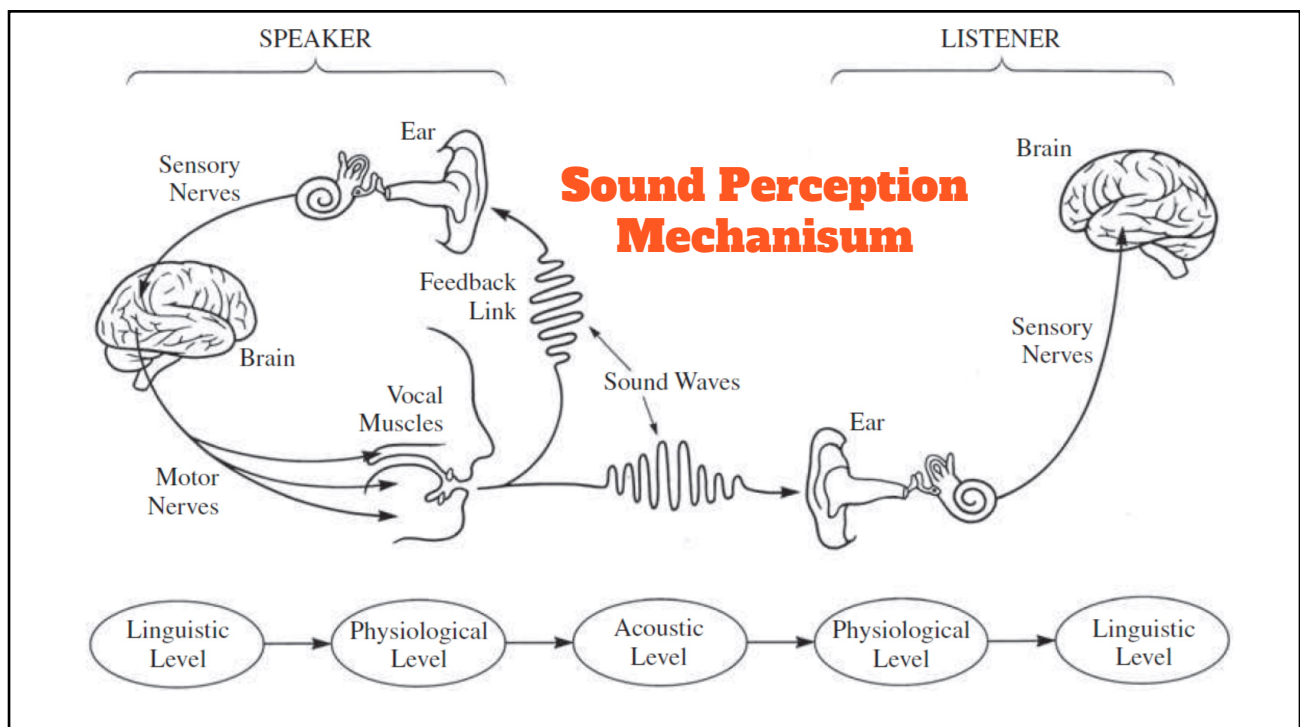
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Example: Different between /b/ and /p/ sounds



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2.2 Vietnamese Phonology and Phonetics

Nguyen Thi Thu Trang

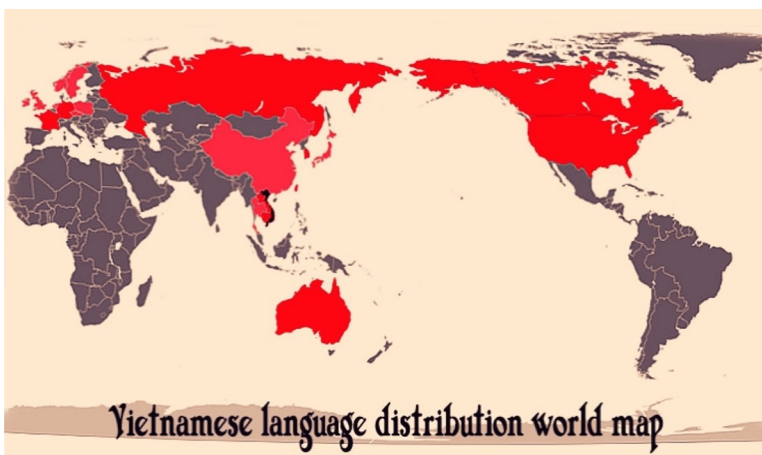
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What is Vietnamese?

- Mon-Khmer sub-branch of the Austroasiatic (Nam Á) language family with mixture of Tai-Kadai (Tày-Thái)
- Official language of Vietnam
- ~100 millions speakers in the world



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Tiếng Việt là ngôn ngữ đơn lập (monosyllabic)

- Tiếng/âm tiết (syllable) được phát âm và viết tách rời
 - Công ty vs. Company
- Từ (word) của tiếng Việt không biến đổi hình thái
 - Tôi **đã** xem phim này rồi.
- Trật tự từ quan trọng
 - Con gà vs. Gà con.

Phát âm	Phân tích	Ghi lại

Tớ ... tên ... là ... Hỏi

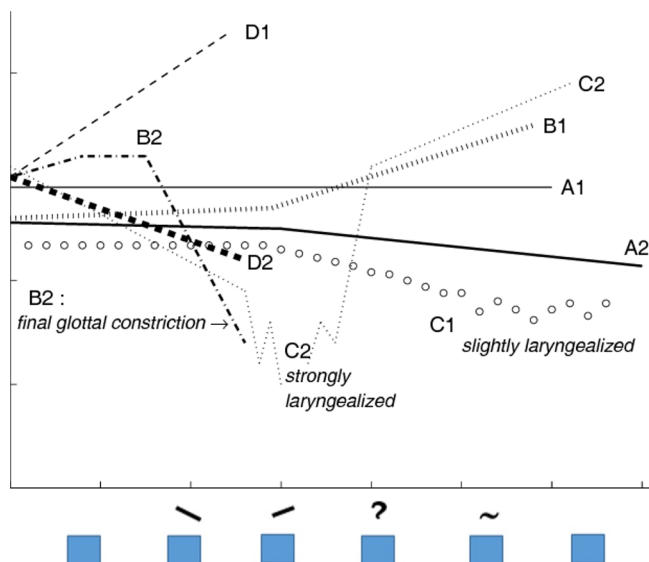
(tớ) (tên) (là) (hỏi)

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Tiếng Việt là ngôn ngữ có thanh điệu (tonal)

- 1 - Ngang (A1)
- 2 - Huyền (A2)
- 3 - Hỏi (C1)
- 4 - Ngã (C2)
- 5a, 5b - Sắc (B1, D1)
- 6a, 6b - Nặng (B2, D2)



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Hanoi Vietnamese¹ tone system (Michaud 2004)

- 6 tones in sonorant-final syllables
- 2 tones in obstruent-final syllables (ending with /p t k/)

#	Name	Duration	Example
1	Level	Long	ba [6a-1] (three)
2	Falling	Long	bà [6a-2] (grand-mother)
3	Curve	Long	bả [6a-3] (poison)
4	Broken	Long	bã [6a-4] (residue)
5a	Rising	Long	bá [6a-5a] (to embrace)
5b	Rising*	Short	bác [6a-5b] (uncle)
6a	Drop	Short	bạ [6a-6a] (to strengthen)
6b	Drop*	Short	bạc [6a-6b] (silver)

¹Standard Vietnamese (Thompson 1987)

Michaud (2004). Michaud Alexis. *Final consonants and glottalization: new perspectives from Hanoi Vietnamese*. Phonetica.

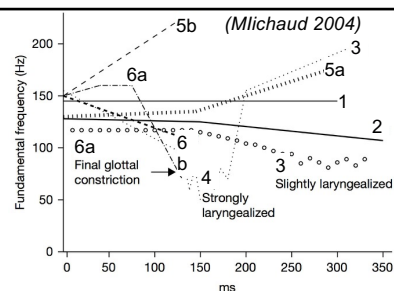
Thompson (1987). Thompson Laurence C. *A Vietnamese Reference Grammar*. University of Hawaii Press.

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Hanoi Vietnamese: A tonal language

Vietnamese tone system [Thompson 1987] [Michaud 2004]

- 6 tones in sonorant-final syllables
- 2 tones in obstruent-final syllables



#	Name	Register	Duration	Pitch contour	Phonation	E.g.
1	Level	High-Mid	Long	Level	Modal	bà (three)
2	Falling	Low	Long	Slightly Falling	Lax	bà (grand-mother)
3	Curve	Low	Long	Falling	Tense	bả (poison)
4	Broken	High	Long	Falling-Rising	Glottal	bã (residue)
5a	Rising	High	Long	Rising	Modal	bá (to embrace)
5b	Rising*	High	Short	Sharply Rising	Breathy	bác (uncle)
6a	Drop	Low	Short	Dropping	Glottal	bạ (to strengthen)
6b	Drop*	Low	Short	Dropping	Breathy	bạc (silver)

Michaud (2004). Michaud Alexis. *Final consonants and glottalization: new perspectives from Hanoi Vietnamese*. *Phonetica*.
Thompson (1987). Thompson Laurence C. *A Vietnamese Reference Grammar*. University of Hawaii Press.

Only different pitch

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Hanoi Vietnamese - G2P Rules for consonants

No.	Graph-eme	Position	Phoneme IPA
1	b	initial only	β
2	đ	initial only	ɗ
3	x, s	initial only	s
4	g, gh	initial only	ɣ
5	kh	initial only	x
6	l	initial only	l
7	v	initial only	v
8	th	initial only	t ^h
9	d, gi, r	initial only	z
10	ph	initial only	f
11	tr, ch	initial only	tʃ
12	h	initial only	h
13	c, k, q	initial only	k

No.	Graph-eme	Position	Phoneme IPA
14	t	initial, final	t
15	p	initial, final	p
16	n	initial, final	n
17	ch	final after i, ê, a	k
18	c	final after u, o, ô	k̚
19	ch, c	final except 17, 18	k
20	m	initial, final	m
21	nh	final after i, ê, a	ɲ
22	nh	initial only	ɲ
23	ng, ngh	initial	ŋ
24	ng	final after u, o, ô	ŋ̚
25	ng	final except 24	ŋ

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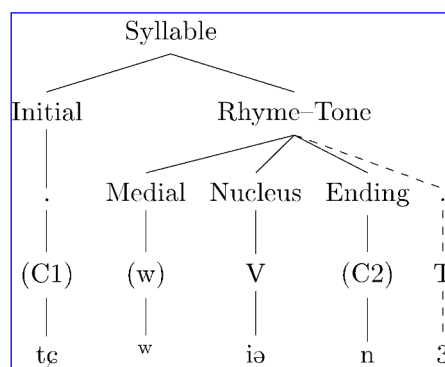
Hanoi Vietnamese - G2P Rules for vowels

Vowel type	Grapheme (Orthography)	P	Vowel type	Grapheme (Orthography)	P
		IPA			IPA
Long vowel	a	a	Short vowel	ă, a (au, ay)	ă
	e	ɛ		â	ɤ
	ê	e		a (anh, ach)	ɛ
	i, y	i		o (ong, oc)	ɔ
	o, oo	ɔ	Diphthong	ia, iê, yê, ya	iə
	ô, ôô	o		ua, uô	uə
	ơ	ɤ		ưa, ươ	uə
	u	u			
	ư	u			

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Vietnamese syllable structure

- Hierarchical structure
(Doan 1977) (Vogel 2004)
 - **Segmental** elements
 - Initial consonant
 - Rhyme (medial, nucleus, ending)
 - **Supra-segmental** element: Lexical Tone (T)
 - mostly covers the rhyme of syllables (Tran 2005)
- Syllabic structure: (C1)(w)V(C2)-T



E.g. chuyển [tɕ^wiən-3] (to move)

Doan (1977). Doan Thien Thuat. *Vietnamese phonetics (in Vietnamese)*. Vacation School and University Publisher.
 Doan (1999). Doan Xuan Kien. *Re-consider a problem of Vietnamese phonetics: Syllable structure (in Vietnamese)*. Hop Luu
 Tran (2005). Tran Do Dat et al. *Influence of F0 on Vietnamese Syllable Perception*. Interspeech 2005. Portugal.
 Vogel (2004). Vogel Irene et al. *Syllable structure in Vietnamese*. TEAL Workshop. Taiwan.

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Structure-based syllable types

INITIAL	RHYME			Examples
	Medial	Nucleus	Ending	
		v		“a” /a-1/ (<i>ah</i>), “ủ” /ɤ-3/ (<i>keep wa</i>)
v		v		“lá” /la-5a/ (<i>leaf</i>), “chờ” /tɕɤ-2/ (<i>wait</i>)
	v	v		“oẹ” /we-6a/ (<i>retch</i>), “uy” /wi-1/ (<i>ui</i>)
		v	v	“ích” /ik-5b/ (<i>helpful</i>), “ây” /ɤ-5a/ (<i>ai</i>)
v	v	v		“loé” /lwe-5a/ (<i>flash</i>), “quá” /kwa-2/ (<i>over</i>)
v		v	v	“trẹo” /tɕew-6a/ (<i>sprain</i>), “nhân” /nɤn-2/ (<i>person</i>)
v	v	v	v	“nhuyễn” /ɲwien-4/ (<i>fine</i>), “soát” /sɔt-2/ (<i>inspect</i>)

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Allophone – []

- A variant of a phoneme
- The allophones of a phoneme form a set of sounds that:
 - Do not change the meaning of a word
 - Are all very similar to one another, and
 - Occur in phonetic contexts different from one another (for example, syllable-initial as opposed to syllable-final).
- The differences among allophones can be stated in terms of phonological rules.

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Luật phiên âm tiếng Việt - Âm đệm

- Duy nhất 01 âm đệm /w/ được biết bằng “o” hoặc “u”
- “o” đứng trước các nguyên âm rộng: a, ă, e
 - VD: toát, hoạ hoằn, loè xoè
- “u” đứng trước các nguyên âm: y, ê, ơ, â
 - VD: huy, thuở, huân, Huế

KHUYÊN /xwiən-1/			
kh /x/	u /w/	yê /iə/	n /n/
	1		

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Exercise

- Phonology differences of Northern, Central and Southern Vietnamese
- Reference: https://www.researchgate.net/profile/Deborah-Hwa-Froelich/publication/278524880_Characteristics_of_Vietnamese_Phonology/links/02e7e530f680f7361f000000/Characteristics-of-Vietnamese-Phonology.pdf

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VD: Luật chính tả tiếng Việt cơ bản

Không tính các từ vay mượn

- Âm đầu **g/gh, ng/ngh, c/k**: Trước âm chính:
 - **Hẹp/hơi hẹp**: **i, ê, e** → **gh, ngh, k**
 - VD: kê, kia, nghe, ghe, nghê, ké, ki bo, nghề nghiệp
 - **Rộng/hơi rộng**: **a, ă, â, o, ô, ơ, u, ư** → **g, ng, c**
 - VD: cá, ngốc, gặt gù, ngừ, có, găm, ngủ, gặt, ngọt ngào
- Âm đầu **q/c/k** (đều có âm là /k/)
 - **q** luôn bao giờ cũng đi với âm đệm **u** để thành qu
 - VD: qua, quanh quẩn (*làm tròn môi âm chính*)
 - **c/k** như trên
- Âm cuối **c/ch** (đều có âm là /k/): Sau âm chính:
 - **Hẹp/rộng**: **i, ê, a** → **ch**, còn lại là “**c**”
 - VD: bích, ốc, ếch, khách, ích, lúc, bác, bức xúc

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Exercise - Luật chính tả / phiên âm

- “gia” và “ghia”; “củ”, “cả”, “cua” và “qua”
- Quốc gia Qata, tỉnh Đắk Lắk



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Discussion - Luật chính tả / phiên âm

- “gia” và “ghia”, “qua” và “quá”

gi		a	
----	--	---	--

c		ua	
---	--	----	--

gh		ia	
----	--	----	--

q	w	a	
---	---	---	--

- Quốc gia Qata, tỉnh Đắk Lắk
→ Từ vay mượn: Qatar và Dak Lak (Darlac)

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Luật chính tả tiếng Việt cơ bản Không áp dụng cho các từ vay mượn

- Âm chính: Nguyên âm y/i
 - Viết là “y”
 - Sau âm đệm “u”: huy, tuy, thúy, quy...
 - Sau nguyên âm ngắn a như: ây, ay
 - Đứng trước ê khi chữ đó không có âm đầu như: yêu, yết, yếm
 - Đứng một mình: y tế, ý nghĩ. Ngoại lệ: í ới
 - Viết “i/y” đều đúng:
 - Không có âm cuối, khuyến khích viết i: Châu Mỹ/Châu Mỹ, Địa lí/Địa lý, Bác sĩ/Bác sỹ
 - Còn lại là “i”
- Exercise
 - “bay” và “bai”, “tuy” và “tui”

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Discussion - Luật chính tả / phiên âm

- “bay” và “bai”, “tuy” và “tui”

b		a	y
/b		ă	j/

t	u	y	
/t	w	i	/

b		a	i
/b		a	j/

t		u	i
/t		u	j/

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Exercise

- Record in Praat for the following sentences:
“Dùng tay vuốt mái tóc bạch kim của em qua tai”
- Analyze and label at allophone level for those sentences

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